Loreto Peter Alonzi III

School of Data Science		
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Professional Experience

- Assistant Professor of Data Science, General Faculty, University of Virginia, 2022 August -present
- Open Data Lab Project Manager and Data Scientist, School of Data Science, University of Virginia, 2018 October - 2022 August
- Senior Research Data Scientist, Research Data Services, University Library, 2014 October - 2018 October

Education

- Postdoctoral Research Associate, University of Washington, Department of Physics, Center for Experimental Nuclear Physics and Astrophysics, 2012 June - 2014 September
- Ph.D. Physics, *University of Virginia*, 2012.
- M.A. Physics, *University of Virginia*, 2007.
- B.S. Physics with Honors et Magna cum Laude, *The College of William and Mary in Virginia*, 2005.

Honors and Awards

- Breakthrough Prize in Fundamental Physics Laureate - $g-2$ collaboration, 2026.
- Excellence in Public Service, UVA Faculty Award, 2026.
- Community Engaged Service Learning Model Proposal, UVA Provost, 2026
- SEDSI Best Paper in Innovation Track, The Future of Data Science Education, 2024
- SIEDS Best Paper in Track, 2024.
- SIEDS Best Paper in Track, 2025.

- University of Virginia Society of Fellows, Junior Fellow, 2010.
- Eagle Scout, BSA NEIC Troop 5, 1999.

Publications and Scholarly Work

Archival Peer Reviewed Journal Articles

- Wright, B., Jones, G., Parker, M., Suen, A., Anand, A., **Alonzi, L.P.**. "The Scientific Method as a Guiding Principle for the Data Science Lifecycle" Harvard Data Science Review (July 2026)
- **Best Paper in Track** B. Wright, **L. P. Alonzi**, A. Rivera. "The Future of Data Science Education" 53rd Annual Southeast Decision Science Institute (March 2024). link, arXiv link
- M.A.McCulloch, **L. P. Alonzi**, S. C. White, F. Haregu, and M. D. Porter. "Pediatric Donor Heart Acceptance Practices in the United States: What Is Really Being Considered?" Pediatric Transplantation 28, no. 1 (November 2023). https://doi.org/10.1111/petr.14649.
- **Top Journal, 1,849 citations** – B. Abi, et al. "Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm" (Muon $g-2$ Collaboration) Phys. Rev. Lett. 126, 141801 (2021)
- J. Grigsby, et al. "Deep learning analysis of deeply virtual exclusive photoproduction" Phys. Rev. D 104, 016001 (2021)
- **Top Journal** – D. Blyth, et al. "First Observation of P-odd γ Asymmetry in Polarized Neutron Capture on Hydrogen" (NPDGamma Collaboration) Phys. Rev. Lett. 121, 242002 (2018)
- D. Počanić, et al. "PEN: a low energy test of lepton universality", *PoS HQL* 2016 (2017) 042
- **L.P.Alonzi**, et al. "The calorimeter system of the new muon $g-2$ experiment at Fermilab", *Nucl.Instrum.Meth. A* 824 (2016) 718-720
- Muon $g-2$ collaboration, "The Measurement of the Anomalous Magnetic Moment of the Muon at Fermilab", *J.Phys.Chem.Ref.Data* 44 (2015) no.3, 031211
- A.T. Fienberg, et al. "Studies of an array of PbF2 Cherenkov crystals with large-area SiPM readout", *Nucl.Instrum.Meth. A* 783 12-21 (2015)
- D. Počanić, et al. "Nab: Measurement Principles, Apparatus and Uncertainties", *Nucl.Instrum.Meth. A* 611, 211-215 (2009).

- D. Počanić(Virginia U.), L.P. Alonzi(Virginia U.), V.A. Baranov(Dubna, JINR), W. Bertl(PSI, Villigen), Yu.M. Bystritsky(Dubna, JINR) et al. (PEN Collaboration), Investigation of Rare Pion Decays with the PIBETA Spectrometer, *Phys.Part.Nucl.Lett. 15 6, 610-620* (2018).
- **Top Journal** – NPDGamma Collaboration, First Observation of P-odd γ Asymmetry in Polarized Neutron Capture on Hydrogen, *Phys.Rev.Lett 121 24 242002* (2018).
- E. Frlez L.P. Alonzi for the PEN Collaboration, Precise Measurement of $\pi^+ \to e^+\nu$ Branching Ratio, *New Trends in High-Energy Physics 2008, 97-106* (2008).

Archival peer reviewed conference proceedings, books, book chapters, and other categories

- **Best Paper in Track** Afi, Z., Atmavilas, S., Bedal, S., Bernard, O., Gupta, M., Lee, C., **Alonzi, L. P., III**, Smith, M. C., White, Vitale M (2025) "High Utilizers of the Albemarle and Charlottesville Criminal Justice Systems", Proceedings of the 2025 IEEE Systems and Information Engineering Design Symposium, Charlottesville, Virginia, (May 2025)
- Lu, I., Seo, R., **Alonzi, L. P., III**, Smith, M. C., White, Vitale M (2025) "Optimizing Local Police Practicies with a Data Visualization Dashboard", Proceedings of the 2025 IEEE Systems and Information Engineering Design Symposium, Charlottesville, Virginia, (May 2025)
- **L.P. Alonzi**, B. Wright, A. Rivera. "Mastermind has Entropy", EAAI-25 Philadelphia, (March 2025).
- **Best Paper in Track** Banino, S., Boulos, G., Craft, C., Phillips, L., Sydnor, S. **Alonzi, L. P., III**, Smith, M. C., White, Vitale M (2024) "Analyzing Candidates for Home Electronic Incarceration on Return-to-Custody Rates for Inmates", Proceedings of the 2024 IEEE Systems and Information Engineering Design Symposium, Charlottesville, Virginia, (April 2024)
- Yik, B. J., **Alonzi III, L. P.**, Wright, B. M., & Stains, M. "Specifications Grading in Data Science Education: Implementation and Evaluation of Proposed Student Outcomes". Presented at the 19th Annual Southeastern STEM Education Research Conference (S2ERC; University of Alabama at Birmingham), Birmingham, AL, January 10–11, 2025 [Conference canceled to due severe weather]. The handbook of presentations and abstracts can be found here: link

- R. Alvarado, **L. P. Alonzi**. "The Virginia Model of Data Science: Humans, Data, Machines" Academic Data Science Alliance Annual Meeting, https://adsa.pubpub.org/proceedings-of-the-2024-adsa-annual-meeting, (October 2024).
- **Alonzi, L.P.**, Yap, Y. "Co-designing with Students: Building a Responsive and Inclusive Syllabus" UVA CTE Pedagogy Summit (May 2024)
- B. Wright, **L. P. Alonzi**, A. Rivera. "What story can you tell from your cell phone apps?: A Data Science Lab Exercise" 53rd Annual Southeast Decision Science Institute (March 2024).
- B. Wright, **L. P. Alonzi**, A. Rivera. "A Better Introduction to Data Science" ADSA Annual Meeting, San Antonio (October 2023).
- **L. P. Alonzi**. "What to Leave in and What to Leave Out?: Teaching Students About Data Collection" ADSA Annual Meeting, San Antonio (October 2023).
- Editor: van Tuyl, S. "Hiring, Managing, and Retaining Data Scientists and Research Software Engineers in Academia: A Career Guidebook from ADSA and US-RSE" ADSA Workshop Chicago (2022), published link (August 2023)
- "Machine Learning for global fitting of CFF from sparse data", Data Science Roadmap to compton form factors of quarks and gluons, Center for Nuclear Femtography, (September 2020).
- "Data Science and Femtography", CNF2019 Symposium, SURA Headquarters, August 2019.
- "The New Muon g-2 Experiment: E989 Status and Progress Update", SESAPS 81st annual meeting, November 2014.
- **L.P. Alonzi** for the PEN Collaboration, "Precision Measurements of Rare Pion Decay Channels", University of Washington, CENPA, March 2012.
- **L.P. Alonzi** for the g-2 Collaboration, "The New Muon g-2 Experiment: E989 Status and Progress Update", *APS NW Section Meeting* (2014).
- **L.P. Alonzi** for the PEN Collaboration, "A Mini Time Projection Chamber of the PEN Experiment", *April APS Meeting* (2010).
- C.J. Glaser for the PEN Collaboration, 'PEN experiment: a precise test of lepton universality", *CIPANP* (2018).
- **L.P. Alonzi** for the Nab Collaboration, "Precision Measurement of a and b in Neutron Beta Decay: The Nab/abBA Experimental Program", *April APS Meeting* (2009).

- D. Počanić for the PEN collaboration, "New Results in Rare Allowed Muon and Pion Decays", *Int.J.Mod.Phys.Conf.Ser. 35 (2014) 1460437, 9th International Workshop on e+ e- Collisions "From phi to psi 2013"* (2014).

- D. Počanić for the PEN collaboration, "New Results in Rare Allowed Muon and Pion Decays", *AIP Conf.Proc. 1560 (2013) 1, 128-130 to CIPANP 2012* (2013).

- S. Baessler for the NAB collaboration, Neutron Beta Decay Studies with Nab, *AIP Conf.Proc. 1560 (2013) 1, 114-116 to CIPANP 2012* (2012).

- D. Kawall for the New Muon g-2 Collaboration, "The New Muon g-2 Experiment at Fermilab", *11th Conference on the Intersections of Particle and Nuclear Physics (2012)*, *AIP Conf.Proc. 1560 (2013) 106-108* (2013)

- S. Baessler for the Nab Collaboration, "Neutron Beta Decay Studies with Nab", *11th Conference on the Intersections of Particle and Nuclear Physics* (2012).

- D. Počanić for the PEN collaboration, "Rare Pion and Muon Decay Measurements: The PIBETA and PEN Experiments", *6th International Workshop on Chiral Dynamics* (2009).

- D. Počanić for the PEN collaboration, "PEN Experiment: A Precise Measurement of the $\pi^+ \to e^+\nu$ Decay Branching Fraction", *10th Conference on the Intersections of Particle and Nuclear Physics conference* (2009).

- A. Palladino for the PEN Collaboration, "Maximum Likelihood Analysis for the PEN Experiment", *April APS Meeting* (2010).

- A. Palladino for the PEN collaboration, "The PEN Experiment at PSI: Testing Lepton Universality", *Seminar on Particle and Astrophysics, Universität Zürich* (2010).

- L. Barron-Palos, et al., "Measurement of parity-violating neutron capture gamma asymmetries at low-energies" XXXII Symposium on Nuclear Physics, Revista Mexicana de Fisica Vol. 55, 2 (2009) 18-22.

  Barron-Palos L, Alarcon R, **Alonzi LP**, B.Lauss, et al. "Measurement of parity-violating neutron capture gamma asymmetries at low-energies" XXXII Symposium on Nuclear Physics, Hotel Hacienda, Morelos, Mexico, January 5-8,2009, Revista Mexicana De Fisica Vol. 55, 2 (2009) 18-22.

- E. Frlež for the PEN collaboration, "Central Particle Tracking Detectors in the PEN Experiment", *Fall APS DNP Meeting* (2008).

- E. Frlež for the PEN collaboration, "Precise Measurement of the $\pi^+ \to e^+\nu$ Branching Ratio", *New Trends in High Energy Physics, Yalta* (2008).

- S. Ritt for the PEN collaboration, "Precise Measurement of the $\pi^+ \to e^+\nu$ Branching Ratio", *Swiss Institute for Particle Physics Annual Meeting (CHIPP)* (2008).
- D. Počanić for the Nab Collaboration, "Nab: Measurement Principles, Apparatus and Uncertainties", *International Workshop for Particle Physics with Slow Neutrons* (2008).
- A. Palladino for the PEN collaboration, "Waveform Analysis for a Precision Pion Decay Measurement", *Fall APS DNP Meeting* (2008).
- D. Počanić for the Nab Collaboration, "The Nab Neutron Decay Correlation Experiment", *American Conference on Neutron Scattering* (2008).
- D. Počanić for the PEN collaboration, "PEN: A Sensitive Search for Non-(V-A) Weak Process", *18th International Conference on Particles and Nuclei* (2008).
- D. Počanić for the PEN collaboration, "Rare Pion and Muon Decays: Summary of Results and Prospects", *Low Energy Precision Electroweak Physics in the LHC Era* (2008).

h-index and citation count

- h-index: 10 – see Profile on inSPIREHEP
- citation count: 2,548 – see Profile on inSPIREHEP

Graduate Students

Doctoral (supervised)

- Michael Murray, University of Washington '17, Current: Lead Data Scientist at OneBridge Solutions
- Matthias Smith, University of Washington '17, Current: ML Scientist at Flexport
- Aaron Feinberg, University of Washington '19, Current: Machine Learning Researcher at BlackBerry
- Brandon Kriesten, University of Virginia '21, Current: Postdoc at Argonne

MSDS

- Isabella Lu, University of Virginia, Graduated MSDS 2026, Current: Student
- Mike Yang, University of Virginia, Graduated MSDS 2019, Current:
- Sai Leela Krishna Prasanth, University of Virginia, Graduated MSDS 2018, Current:

Undergraduate Students

- Kazimir Wall, University of Washington, $g-2$
- Grace Tong, UVA. $SIWIF$
- Jake Grigsby, UVA. $SIWIF$
- Philip Velie, UVA. $SIWIF$
- Emma Yeats, UVA. $SIWIF$
- Fernanda Yepez Lopes, UVA. $SIWIF$
- Aditi Jain, UVA, $EBDM$
- Amelia Norman, UVA. $EBDM$
- Michela Nardi, UVA. $EBDM$
- Carter Levinson, UVA. $EBDM$
- Meredith Lawrence, UVA. $EBDM$
- Audrey Kamauff, UVA. $EBDM$
- James Wilby, UVA. $EBDM$
- William Burge, UVA. $EBDM$
- Paul Dalton, UVA. $EBDM$
- Kathryn Murray, UVA. $EBDM$
- Thomas Owen, UVA. $EBDM$
- Charles Rowe, UVA. $EBDM$
- Ashwin Sundaram, UVA. $EBDM$
- Adam Will , UVA. $EBDM$
- Emma Boland, UVA. $EBDM$

- Caroline O'Brien, UVA. *EBDM*
- John Henry Oliphant, UVA. *EBDM*
- Josh Williams, UVA. *EBDM*
- Henry Bramham, UVA. *EBDM*
- Claire Deaver, UVA. *EBDM*
- Sean Domnick, UVA. *EBDM*
- Emma Hand, UVA. *EBDM*
- Emily Ledwith, UVA. *EBDM*
- Noah O'Neill, UVA. *EBDM*
- Callie Weiler, UVA. *EBDM*
- Bella Lu, UVA, *EBDM*, https://orcid.org/0009-0007-6152-072X
- Grace Boland, UVA. *EBDM*
- Colin Cool, UVA. *EBDM*
- Nathaniel Donkoh-Moore, UVA. *EBDM*
- Patrick Leonard, UVA. *EBDM*
- Maddie McNult, UVA. *EBDM*
- George Corbin, UVA. *EBDM*
- Nora Dale, UVA. *EBDM*
- Aatmika Deshpande, UVA. *EBDM*
- Katherine Korngiebel, UVA. *EBDM*
- Paige Krablin, UVA. *EBDM*
- Emma Wilt, UVA. *EBDM*
- Stella Banino, UVA. *EBDM*
- Chris Craft, UVA. *EBDM*
- Laura Phillips, UVA. *EBDM*
- Sally Sydnor, UVA. *EBDM*
- George Boulos, UVA. *EBDM*
- Josh Dornfeld, UVA. *EBDM*

- Imani Hankinson, UVA. $EBDM$
- Livia Hughes, UVA. $EBDM$
- Sarah Murphy, UVA. $EBDM$
- Ronica Peraka, UVA. $EBDM$
- McBride Rawson, UVA. $EBDM$
- Zak Afi, UVA. $CJDTI$
- Sudarshan Atmavilas, UVA. $CJDTI$
- Sarah Bedal, UVA. $CJDTI$
- Olivia Bernard, UVA. $CJDTI$
- Mohini Gupta, UVA. $CJDTI$
- Caroline Lee, UVA. $CJDTI$
- Isabella Lu, UVA. $CJDTI$
- Rachel Seo, UVA. $CJDTI$
- Jonas Lee, UVA. $CJDTI$
- Saiesha Gohil, UVA. $CJDTI$
- Mitchell Palmer, UVA. $CJDTI$
- Jay Sorkin, UVA. $CJDTI$
- Kevin Villalobos, UVA. $CJDTI$
- Ryan Yates, UVA. $CJDTI$

Funded and Under Review Grant Proposals as PI/Co-PI

- [**Proposal under review**] Optimization of the CEBAF Polarized Electron Beam for the Systematic Control of the MOLLER Experiment at Jefferson Laboratory (2026) $749,906
  - (Senior personnel) **L.P.Alonzi**
  - Department of Energy (DE-FOA-0003612)
- [**LOI submitted**] Criminal Justice Data Integration Training & Outreach (2026) $50,000
  - (PI) **L.P.Alonzi**

    - Center for Public Safety and Justice (CPSJ)/Howard and Carol Safir Foundation (HCSF) Public Safety Initiative

- CJDTI Partner Support (2025) $3,000
    - (PI) **L.P.Alonzi**
    - Offender Aid and Restoration (OAR)
- Community Engaged Service Learning Grant (2026) $7,000
    - (PI) **L.P.Alonzi**
    - UVA Office of Academic Outreach
    - For: SYS 4054 Sp26
- Community Engaged Service Learning Grant (2025) $7,000
    - (PI) **L.P.Alonzi**
    - UVA Office of Academic Outreach
    - For: SYS 4053 Fa25
- Criminal Justice Data Training Initiative (2025) $78,594
    - (PI) **L.P.Alonzi**
    - The Jefferson Trust
- Building Inclusive Student-Informed Courses (2023) $9,600
    - (PI) R. Schmidt; (Co-PI): **L.P.Alonzi**
    - Inclusive Excellence award from UVA Division for Diversity, Equity, and Inclusion
- Data Science Active Learning Lab (2022) $150,000
    - (PI) B.Wright; (Co-PI): **L.P.Alonzi**
    - President and Provost's Fund for Institutionally Related Research
- Optimizing pediatric donor heart utilization using big data analytics (2022) $133,078
    - (PI) M.McCulloch; (Co-PI): **L.P.Alonzi**, M. Porter
    - The Jefferson Trust
- SIWIF: Summer Institute on Wigner Imaging and Femtography (2019) $50,000
    - (PI) S.Liuti; (Co-PIs): **L.P.Alonzi**, M.Burkhardt, D.Keller, O.Pfister
    - Southeastern Universities Research Association, Inc. Grant Agreement No. C2020-FEMT-006-05 (continuation supplemental)

- SIWIF: Summer Institute on Wigner Imaging and Femtography (2019) $30,000
  - (PI) S.Liuti; (Co-PIs): **L.P.Alonzi**, M.Burkhardt, D.Keller, O.Pfister
  - Southeastern Universities Research Association, Inc. Grant Agreement No. C2020-FEMT-006-05
- Wigner Imaging (2018) $50,000
  - (PI) S.Liuti; (Co-PIs): **L.P.Alonzi**, M.Burkhardt, D.Keller, O.Pfister
  - Southeastern Universities Research Association, Inc. Grant Agreement No. C2019-FEMT-002-04

Presentations, Invited

- "Introduction to R", DAACS Open Academy, Fall 2021
- "What Entropy and Impedance Mean in Data Science", Jefferson Lab AI Lunch Series, November 2020
- "Machine Learning for global fitting of CFF from sparse data", Data Science Roadmap to compton form factors of quarks and gluons, Center for Nuclear Femtography, September 2020.
- "Data Science and Femtography", CNF2019 Symposium, SURA Headquarters, August 2019.
- "Intro to Data Science", CSInstitute, July 2019.
- "The New Muon g-2 Experiment: E989 Status and Progress Update", SESAPS 81st annual meeting, November 2014.
- "Using the Library to Advance Scientific Data Practices", University of Virginia, July 2014.
- L.P. Alonzi for the PEN Collaboration, "Precision Measurements of Rare Pion Decay Channels", University of Washington, CENPA, March 2012.

Internal Service

- Undergraduate Committee, 2020-present, (School)
- Academic Committee, 2023-present, (School)
- Deputy Director of Undergraduate Programs, 2024-present (School)
- Darden Collaboratory Steering Committee, 2022-present (School)
- ADSTP Hiring Committee, 2023, (School)

Professional Service

- Educational Advances in Artificial Intelligence, Program Committee, 2026
- Academic Data Science Alliance, Annual Meeting, Program Committee, 2024
- Innovations in Pedagogy Summit, Center for Teaching Excellence, Program Committee, 2023

ORCID

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Teaching Experience

Full record see: github.com/alonzi/cv

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